AGROSPEAR CROPCARE CO., LTD.

STANDARD OPERATING PROCEDURE: SC & WDG MANUFACTURING & QC

悬浮剂 (SC) 与 水分散粒剂 (WDG) 生产制程与质量控制标准操作规程

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Compliance: FAO/WHO Specs & CIPAC Methods
Target Formulation: SC (Suspension Concentrate) / WDG

1. PURPOSE & SCOPE / 目的与适用范围

To standardize the production parameters, in-process quality control (IPQC), and final quality assurance (FQC) standards for Suspension Concentrate (SC) and Water Dispersible Granule (WDG) pesticide formulations produced by AGROSPEAR CROPCARE CO., LTD.

规范 AGROSPEAR CROPCARE CO., LTD. 旗下悬浮剂（SC）及水分散粒剂（WDG）农药剂型的工业化生产操作、制程质量监控（IPQC）及成品出厂检验（FQC）全流程，确保批次稳定性与国际农药标准（FAO/WHO/CIPAC）合规。

2. SUSPENSION CONCENTRATE (SC) SOP / 悬浮剂生产与质控标准规程

2.1 SC Manufacturing Process Workflow / 悬浮剂生产工序控制

Process Step / 工序	Operational Standard / 操作标准 (EN & CN)	Critical Control Points (CCP) / 关键参数	Frequency / 频次
Step 1: Pre-dispersion **预混合搅拌**	**Charge water, wetting/dispersing agents, antifreeze (e.g., glycol), and defoamer. High-shear disperse active ingredient (AI) powder at 800–1200 RPM for 30–45 min.** 按配方加入水、润湿分散剂、防冻剂及消泡剂。投入原药粉末，在 800–1200 RPM 下高速剪切混合 30–45 分钟，形成均相初浆料。	• Temperature < 40°C • Uniform slurry without lumps • 温度控制 < 40°C，无团块	Every Batch 每批次
Step 2: Wet Bead Milling **卧式砂磨研磨**	**Circulate slurry through horizontal sand mills (0.8–1.0mm Yttrium-stabilized Zirconia beads). Monitor chill-water jacket to prevent thermal degradation of active ingredients.** 经由氧化锆珠砂磨机进行两段研磨。严密监控冷却水夹套，防止磨腔过热导致原药分解或晶型转化。	• Outlet Temp ≤ 38°C • Mill Chamber Pressure < 0.25 MPa • **Particle Size: D50 < 2.0μm, D90 < 5.0μm**	Every 45 min per mill 每磨机 45 分钟取样
Step 3: Post-Addition & Stabilization **调配与结构增稠**	**Transfer milled base to blending vessel. Add pre-hydrated structuring agents (Xanthan Gum + Veegum) and biocide (BIT/MIT). Agitate at low shear (200–400 RPM) for 60 min.** 物料导入调配釜，加入预水化黄原胶-硅酸镁铝复合增稠胶体及防腐防霉剂，低速剪切搅拌 60 分钟以构建抗分层网络结构。	• Viscosity: 400–1200 mPa·s • Thixotropic index ≥ 2.5 • 黏度与触变性验证	Post-mixing 调配完成后
Step 4: Filtration & Packaging **精细过滤与灌装**	**Filter final product through 100-mesh stainless steel sieve. Automate volumetric filling, inductive foil sealing, and laser batch coding.** 经 100 目不锈钢袋式过滤器滤除杂质。进入自动化灌装线，实行电磁感应铝箔封口与激光喷码。	• Net Weight / Volume accuracy ±0.5% • Foil seal 100% leak-proof check • 净含量偏差与铝箔封口密闭性	Continuous / Every 30 min 每 30 分钟抽检

2.2 SC Quality Control Specification (FQC) / 悬浮剂成品出厂质检指标

Test Item / 检测项目	Standard Specification / 指标要求	Testing Method / 检测方法标准	Evaluation / 判定准则
Active Ingredient Content (含量)	Nominal content ± 5.0% (标称含量偏差范围内)	HPLC (In-house / CIPAC MT)	Mandatory / 必检项
Suspensibility (悬浮率)	≥ 90.0 % (before & after accelerated storage)	CIPAC MT 184 (Standard Water D, 30°C)	Mandatory / 必检项
Particle Size Distribution (粒度)	D50 ≤ 2.0 μm, D90 ≤ 5.0 μm	Laser Diffraction (Malvern Mastersizer)	Mandatory / 必检项
pH Range (酸碱度)	5.0 – 8.5 (1% aqueous emulsion)	CIPAC MT 75.3 (pH Electrode Meter)	Mandatory / 必检项
Pourability / Residue (倾倒性)	Residue ≤ 5.0 %, Rinsed residue ≤ 0.8 %	CIPAC MT 148.1	Audit Check / 抽检
Persistent Foaming (持久起泡性)	Foam volume ≤ 60 mL after 1 min	CIPAC MT 47.2	Mandatory / 必检项
Accelerated Heat Stability (热储)	54 ± 2°C for 14 days, AI degradation ≤ 5.0%	CIPAC MT 46.3 (High Temp Chamber)	Batch Validation / 留样验证

3. WATER DISPERSIBLE GRANULE (WDG) SOP / 水分散粒剂生产与质控标准规程

3.1 WDG Manufacturing Process Workflow / 水分散粒剂生产工序控制

Process Step / 工序	Operational Standard / 操作标准 (EN & CN)	Critical Control Points (CCP) / 关键参数	Frequency / 频次
Step 1: Dry Milling & Blending **气流粉碎与干混**	**Co-blend TC powder, binders (e.g., lignosulfonates), disintegrants (e.g., sulfates), and inorganic carriers. Micronize using Fluidized Jet Mill to achieve D90 < 10 μm.** 将原药、木质素磺酸盐分散剂、硫酸盐崩解剂及载体配料预混。采用超音速气流粉碎机细化至 D90 < 10 μm。	• Jet Mill Inlet Pressure: 0.7–0.8 MPa • Moisture Content < 1.5% • Fineness: Wet sieve 325 mesh ≥ 98%	Every 60 min 每小时 1 次
Step 2: Kneading & Extrusion **湿法捏合与挤压造粒**	**Add 12–18% water into high-speed paddle kneader to form homogeneous dough. Feed into screw/basket extruder with 0.8–1.2mm die plates.** 在捏合机中喷入 12–18% 纯水捏合成团状料。送入螺杆/旋转挤压造粒机，通过 0.8–1.2mm 孔径筛网挤出均匀圆柱状颗粒。	• Kneading Moisture: 14 ± 1.5% • Extrusion Temp ≤ 35°C (no melting) • Die head wear check	Continuous monitoring 连续监控
Step 3: Fluidized Bed Drying **流化床烘干与冷却**	**Dry wet granules in vibrating fluidized bed. Section 1 (hot air drying 70–85°C), Section 2 (cooling section < 30°C). Prevent granule crusting or thermal decomposition.** 挤出颗粒进入振动流化床。一区热风烘干（70–85°C），二区冷风降温（< 30°C）。严格避免高温造成原药熔融结块或分解。	• Inlet Air Temp: 75–85°C • Bed Material Temp < 50°C • **Final Residual Moisture ≤ 2.0%**	Every 30 min 每 30 分钟快速水分测定
Step 4: Sieving & De-dusting **筛分分级与除尘包装**	**Classify through vibrating screen (10–40 mesh / 0.45–2.0 mm). Recycle over-sized (crush) and fines back to kneader. Seal in moisture-proof foil bags.** 通过 10–40 目双层振动筛分出合格颗粒（0.45–2.0 mm）。大颗粒粉碎回流，筛下细粉返回捏合段。自动充氮铝箔袋密封。	• Dust content < 1.0% (CIPAC MT 171) • Granule Appearance: Smooth, non-caking • Sealing integrity	Every batch / 1h check 每批次 / 巡检

3.2 WDG Quality Control Specification (FQC) / 水分散粒剂成品出厂质检指标

Test Item / 检测项目	Standard Specification / 指标要求	Testing Method / 检测方法标准	Evaluation / 判定准则
Active Ingredient Content (含量)	Nominal content ± 5.0% (符合标准规定)	HPLC / GC Analysis	Mandatory / 必检项
Disintegration Time (崩解时间)	≤ 90 seconds (Complete spontaneous dispersion)	CIPAC MT 174 (Inversion cylinder test)	Mandatory / 必检项
Suspensibility (悬浮率)	≥ 85.0 % (at 30°C in CIPAC Water D)	CIPAC MT 184	Mandatory / 必检项
Residual Moisture (残余水分)	≤ 2.0 % (w/w)	Karl Fischer / Halogen Moisture Analyzer	Mandatory / 必检项
Wet Sieve Test (湿筛试验)	≥ 98.0 % through 75 μm (200 mesh)	CIPAC MT 185	Mandatory / 必检项
Attrition Resistance (耐磨碎率)	Hardness / Attrition resistance ≥ 90.0 %	CIPAC MT 178.2	Audit Check / 抽检
Dustiness / Friability (粉尘性)	< 12 mg collected dust (Categorized as nearly dust-free)	CIPAC MT 171	Batch Validation / 抽检

4. OOS INVESTIGATION & BATCH RELEASE / 超标调查与成品放行授权

OUT OF SPECIFICATION (OOS) PROTOCOL: Any test failure triggers immediate quarantine of the batch. Production engineering and QA conduct root-cause analysis (milling time, temperature spike, carrier variance). Rework or disposal must be signed off by the QA Director.
超标（OOS）处置规程： 任何指标不合格须立即隔离整批产品并张贴红卡。由生产工程部与质量保证部联合调查根本原因（砂磨时间、温升异常、助剂相容性等）。返工或报废方案必须经 QA 总监书面签署批准。

Role / 岗位	Designee / 责任人	Signature / 签字	Date / 日期
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